

**SEMINAR SUMMARY:  
JANUARY 12, 2026 - DAVID CHAN  
WHAT IS RESEARCH?**

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*What is Research?* The process of learning enough information to sufficiently answer a question.

In Tuesday January 12th's seminar, David Chan opened the talk with a summary of the course generally. We began by noting that every summary document assignment (due the Friday following the seminar) should contain the following elements:

- the name of the presenter,
- the presentation title, and
- a thorough but concise summary.

The class expectations follow as:

- be on time,
- pay attention,
- ask questions.

The goals for the course in general follow as:

- to discover the research interests of a collection of faculty members,
- to experience a fairly broad range of research topics,
- to experience a broad range of presentation styles, and finally
- to gain a better understanding of what research is and how to do it.

When doing guided research with a professor there are a few expectations that go along with the hours obtained. The following give a fairly comprehensive set of expectations for research operation:

- (1) work consistently,
- (2) meeting regularly,
- (3) explain your results to others (for example in seminars),
- (4) training (e.g. lab policies, software),
- (5) reading,
- (6) getting frustrated a lot.

Some general advice follows as:

- pick something that you are passionate about,

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*Date:* January 15, 2026.

- try to find a professor that you are comfortable with,
- pick something that is in an area of strength
- **think on your own**
- enjoy yourself!

After this description of the courses logistics and goals Dr. Chan got into his individual areas of research, which we'll address in the following sections.

### 1. PROJECT 1. RIPPLED DYNAMICS

The idea at hand here is to study these sequence generators mainly for the purpose of modeling drug interactions with epilepsy patients. Though the lens through which we are looking at these drug interactions is difference equations in the discrete dynamical system field. The curiosity with regards to these sequence generating functions is that they are both chaotic and periodic with their behaviour.

The model that we were given follows as

$$x_{n+1} = \max \left\{ \frac{A_n + D_n H_\epsilon(x_n)}{x_n}, \frac{B_n}{x_n - 1} \right\}$$

where  $A_n$  and  $D_n$  are periodic generating sequences,  $B_n$  is non-periodic, and  $H_\epsilon(x)$  is the heavyside function. We generally simplify the model to:

$$x_{n+1} = \frac{A_n + D_n H_\epsilon(x_n)}{x_n}$$

We then proceeded to go over graphs of these models showing periodic behaviour, and then with persistent iterations mild non-periodic behaviour develops over top of the persistent periodic behaviour.

### 2. PROJECT 2. BARRIER ISLAND EVOLUTION

Barrier islands, take for example Parramore Island on the Eastern Shore of Virginia, have quite a subtly non-stable landmass. So we're attempting to model the island topography over time with regards to both physical and biological stimuli (e.g. tides, storms, wave mechanics, plants, grasses). Specifically noted as the modeling mechanics were Cellular Models/ABM. Furthermore, the mechanics of the models heavily rely upon linear algebra. The final goal with the project would be to eventually develop a classification system upon which a risk metric would be developed (in the same sense as the risk metric for hurricanes).

### 3. PROJECT 3. CORTISOL MODELING

Using ordinary differential equations the goal is to create a model of the hormone cortisol in the bloodstream. The motivation for such a model is that it is well known in the medical community that cortisol is the hormone responsible for stress. Further, we have the goal

here of modeling epinephrine and norepinephrine in the body to measure patient status in the hospital. From what I could tell, these models are in fairly late stages and the work is in the fine-tuning them. Lastly, there is a goal with the work to potentially implement the models in devices that would monitor these hormonal levels and process the models for patient classification for the potential of alerting or automated intervention. Granted, it sounded like the clinical research is a bit off and the ODE work is primary.

#### 4. PROJECT 4. HOSPITAL EPIDEMIOLOGY

This seemed like a fairly expansive research topic using a variety of means at getting traction on different mechanisms of disease spread. The different means of modeling listed were: ODEs, Markov chains, ABM, graphs, and more. Some of the problems that we are gain traction on are:

- disease transmission via bathing wipes
- patterns of disease spread with COVID where there is a comorbidity of chlamydia
- transmission of *Clostridioides difficile* (a highly contagious pathogen via contact)

#### 5. PROJECTS 5, 6, 7 - HUMAN BEHAVIOR

The first and most emphasized point here was that humans are extraordinarily complex, but also that humans are quite interesting to study. Options for research in this area varied from data collection, to statistical modeling to using graph theory to gain traction on the problems.